Reason about equivalent expressions

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: __________________

1. Read the model: Are 7 x (20 + 3) and 7 x 20 + 7 x 3 equivalent? Copy or complete the first step.

2. Which is greater: 100 - 8 x 9 or (100 - 8) x 9?

3. Insert operations between 6, 5 and 4 to make 26.

4. Miro says: Miro thinks changing brackets never changes an answer. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Are 7 x (20 + 3) and 7 x 20 + 7 x 3 equivalent? Copy or complete the first step.
   Answer Yes, both equal 161.

2. Which is greater: 100 - 8 x 9 or (100 - 8) x 9?
   Answer (100 - 8) x 9 is greater: 828 compared with 28.

3. Insert operations between 6, 5 and 4 to make 26.
   Answer 6 x 5 - 4 = 26.

4. Miro says: Miro thinks changing brackets never changes an answer. Is this correct? Explain.
   Answer Brackets can change which values combine first, producing a different result.

Reason about equivalent expressions

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: ________________

1. Which is greater: 100 - 8 x 9 or (100 - 8) x 9?

2. Insert operations between 6, 5 and 4 to make 26.

3. Find three expressions with answer 50 using at least two different operations.

4. Identify and correct this misconception: Miro thinks changing brackets never changes an answer.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. Which is greater: 100 - 8 x 9 or (100 - 8) x 9?
   Answer (100 - 8) x 9 is greater: 828 compared with 28.

2. Insert operations between 6, 5 and 4 to make 26.
   Answer 6 x 5 - 4 = 26.

3. Find three expressions with answer 50 using at least two different operations.
   Answer Answers vary, for example 8 x 6 + 2; 100 / 2; (15 - 5) x 5.

4. Identify and correct this misconception: Miro thinks changing brackets never changes an answer.
   Answer Brackets can change which values combine first, producing a different result.

5. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Reason about equivalent expressions

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: _________________

1. Find three expressions with answer 50 using at least two different operations.

2. Explain why this reasoning fails: Miro thinks changing brackets never changes an answer.

3. Create a new example that tests the same idea as: Insert operations between 6, 5 and 4 to make 26.

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. Find three expressions with answer 50 using at least two different operations.
   Answer Answers vary, for example 8 x 6 + 2; 100 / 2; (15 - 5) x 5.

2. Explain why this reasoning fails: Miro thinks changing brackets never changes an answer.
   Answer Brackets can change which values combine first, producing a different result.

3. Create a new example that tests the same idea as: Insert operations between 6, 5 and 4 to make 26.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

4. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

5. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.